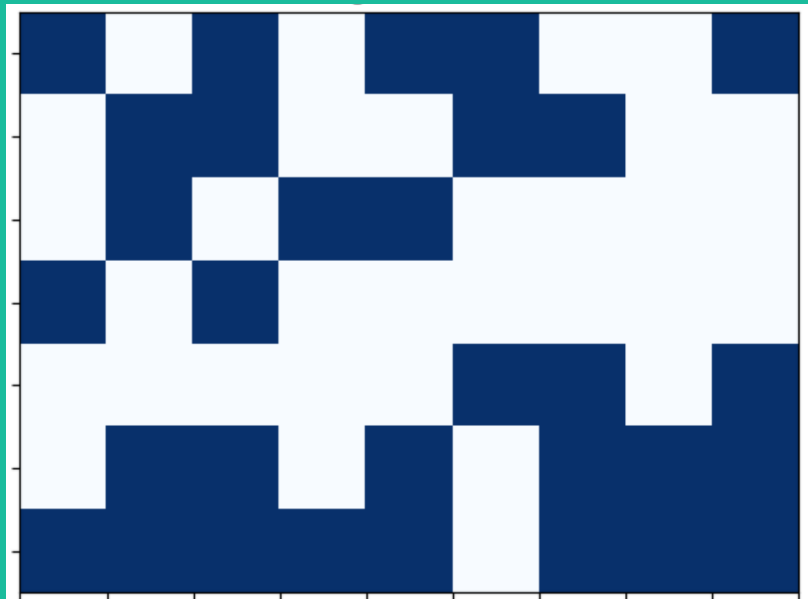




Final Assignment: Economic Complexity Report

Group 4

PART 1: Descriptive Statistics (1)

- Let's start with the input data

```
# Define matrix, connecting (imaginary) countries and activities.
```

```
countries=['Vale of Arryn','The North','The Riverlands', 'Dorne', 'The Stormland','The Westerlands','The Reach']  
activities=[f'P{j}' for j in range(9)]
```

```
matrix = [[1, 0, 1, 0, 1, 1, 0, 0, 1],  
          [0, 1, 1, 0, 0, 1, 1, 0, 0],  
          [0, 1, 0, 1, 1, 0, 0, 0, 0],  
          [1, 0, 1, 0, 0, 0, 0, 0, 0],  
          [0, 0, 0, 0, 0, 1, 1, 0, 1],  
          [0, 1, 1, 0, 1, 0, 1, 1, 1],  
          [1, 1, 1, 1, 1, 0, 1, 1, 1]]
```

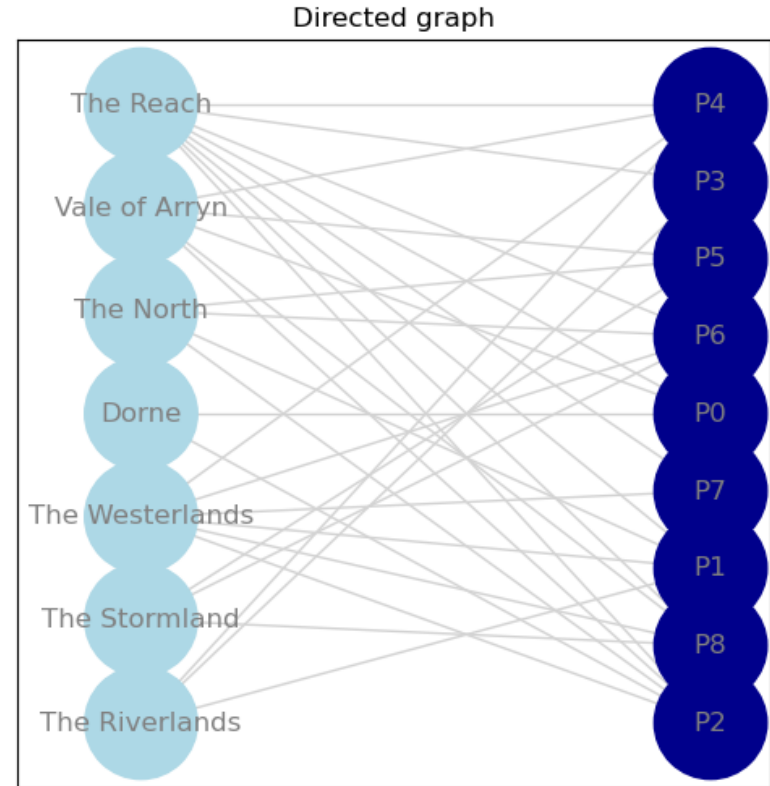
PART 1: Descriptive Statistics (2)

Original countries:

- index ' 0' - Vale of Arryn
- index ' 1' - The North
- index ' 2' - The Riverlands
- index ' 3' - Dorne
- index ' 4' - The Stormland
- index ' 5' - The Westerlands
- index ' 6' - The Reach

List of activities:

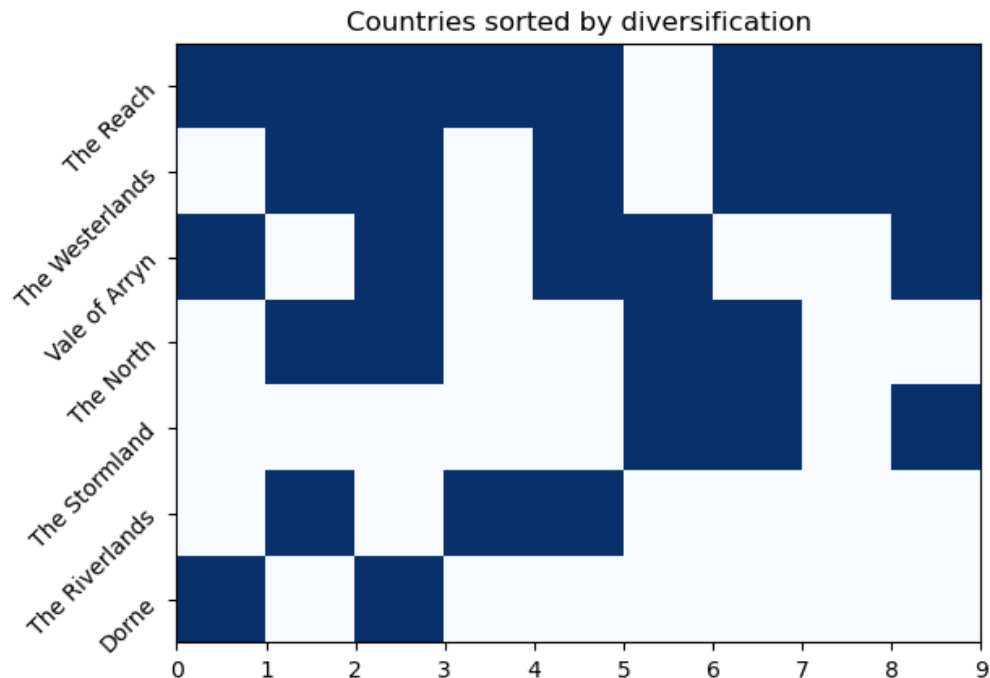
- index ' 0' - P0
- index ' 1' - P1
- index ' 2' - P2
- index ' 3' - P3
- index ' 4' - P4
- index ' 5' - P5
- index ' 6' - P6
- index ' 7' - P7
- index ' 8' - P8



PART 1: Descriptive Statistics (3)

Countries sorted by diversification (Most diverse at the top):

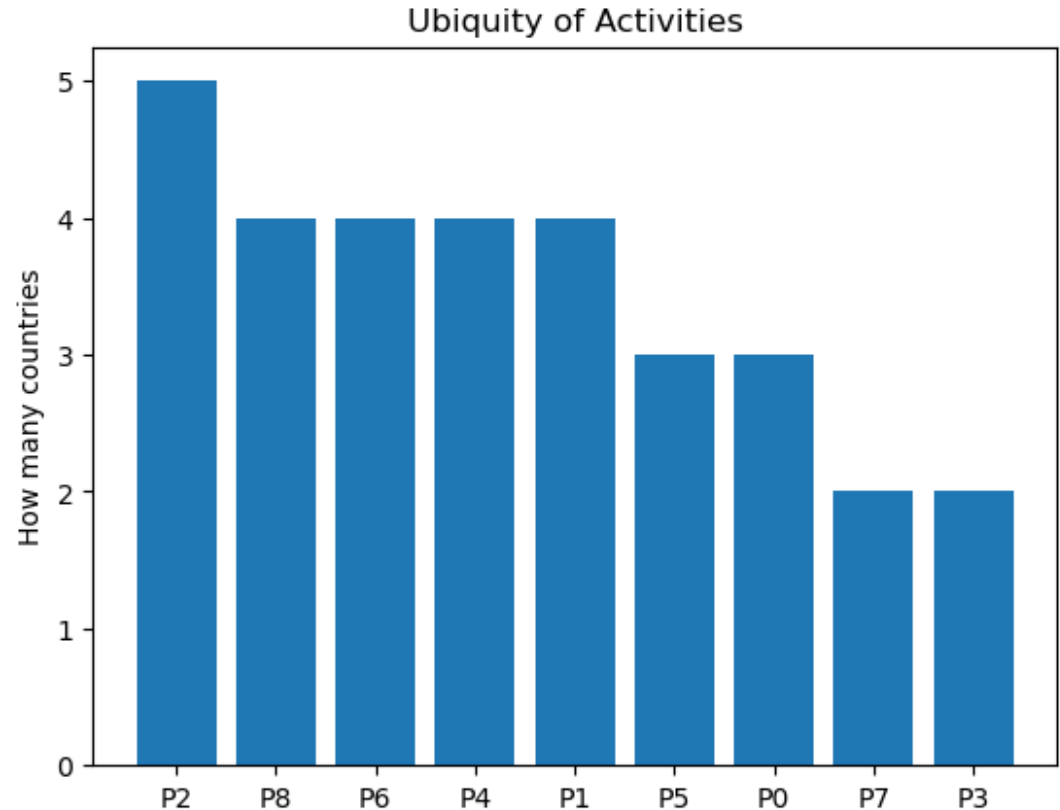
1. - The Reach (diversification: 8)
2. - The Westerlands (diversification: 6)
3. - Vale of Arryn (diversification: 5)
4. - The North (diversification: 4)
5. - The Stormland (diversification: 3)
6. - The Riverlands (diversification: 3)
7. - Dorne (diversification: 2)



PART 1: Descriptive Statistics (4)

- Ubiquity of Products / Activities
("Specialization"):

The most common product is **P2** as five countries (out of 7) are able to produce it; while **P7** and **P3** are the most rare to produce as only two countries are able to do so – either due to complexity or the product is too niche in other countries.



PART 2: Economic Complexity & Fitness Measures (1)

- ECI with eigenvalues

The ECI (Economic Complexity Index) was calculated using the eigenvalues and countries were reordered from the most developed to the least.

(There are for comparison also values with sign removed)

ECI as an eigenvalue problem:

0: Vale of Arryn	= -0.56136
1: The North	= -0.48151
2: The Riverlands	= 2.0548
3: Dorne	= -0.6958
4: The Stormland	= -1.1651
5: The Westerlands	= 0.30489
6: The Reach	= 0.54408

Ordered:

1. - (2) The Riverlands	= 2.0548
2. - (6) The Reach	= 0.54408
3. - (5) The Westerlands	= 0.30489
4. - (1) The North	= -0.48151
5. - (0) Vale of Arryn	= -0.56136
6. - (3) Dorne	= -0.6958
7. - (4) The Stormland	= -1.1651

Ordered (sign removed):

1. - (2) The Riverlands	= 2.0548
2. - (4) The Stormland	= 1.1651
3. - (3) Dorne	= 0.6958
4. - (0) Vale of Arryn	= 0.56136
5. - (6) The Reach	= 0.54408
6. - (1) The North	= 0.48151
7. - (5) The Westerlands	= 0.30489

PART 2: Economic Complexity & Fitness Measures (2)

- **PCI with eigenvalues**

The PCI (Product Complexity Index) was calculated using the eigenvalues and activities were reordered from the most sophisticated product to the least.

- Notice that **P3**, **P1** and **P4** are the most complex which are all produced by The Riverlands (first in ECI ranking).

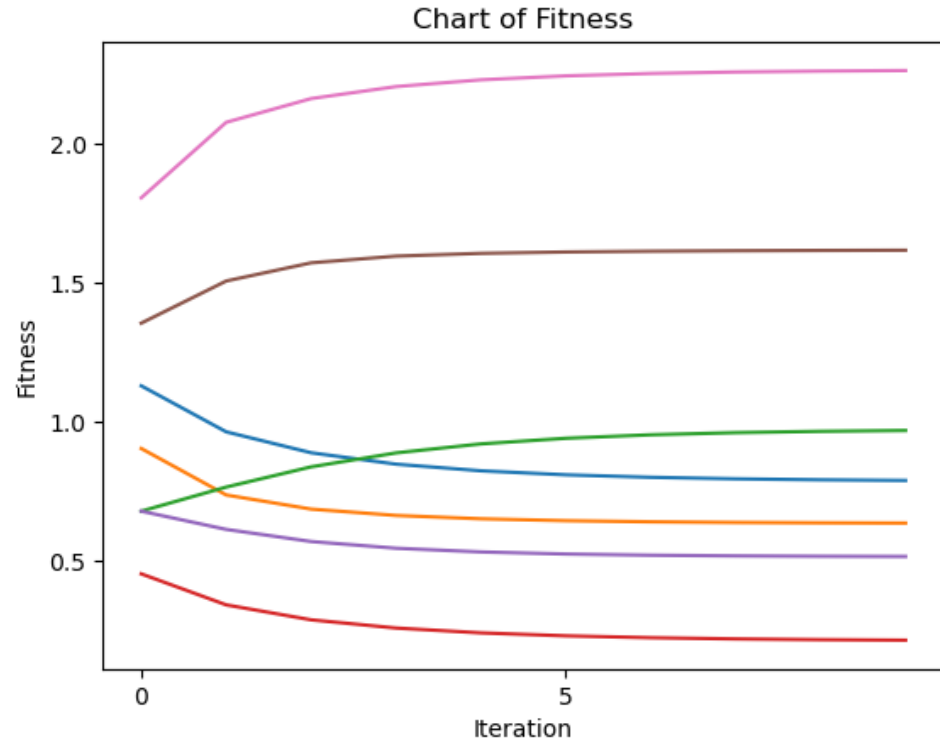
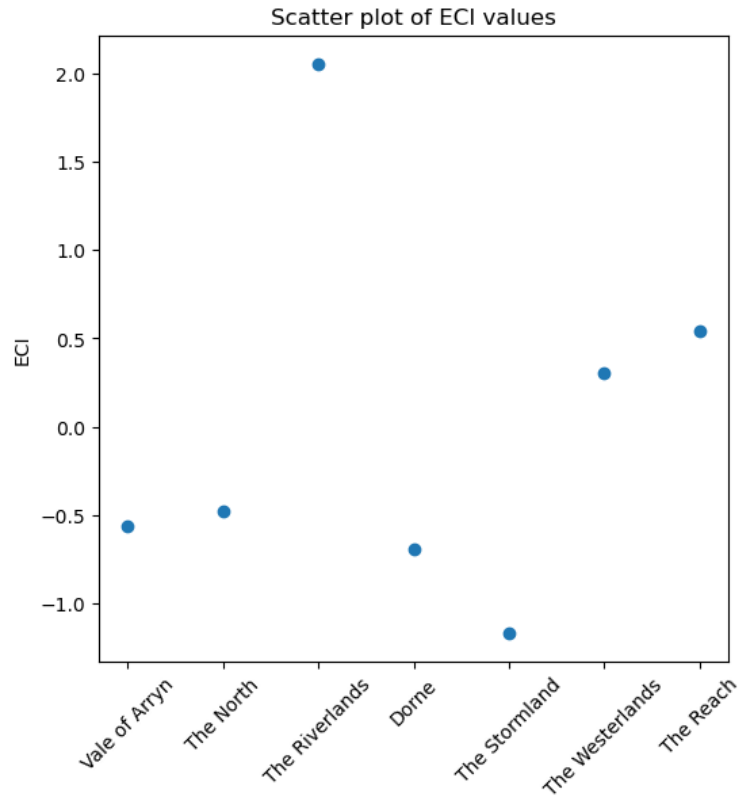
PCI as an eigenvalue problem:

0: P0 = -0.66008
1: P1 = 0.77784
2: P2 = -0.55819
3: P3 = 1.961
4: P4 = 0.7438
5: P5 = -1.5098
6: P6 = -0.59482
7: P7 = 0.46905
8: P8 = -0.62886

Ordered:

1. - (3) P3 = 1.961
2. - (1) P1 = 0.77784
3. - (4) P4 = 0.7438
4. - (7) P7 = 0.46905
5. - (2) P2 = -0.55819
6. - (6) P6 = -0.59482
7. - (8) P8 = -0.62886
8. - (0) P0 = -0.66008
9. - (5) P5 = -1.5098

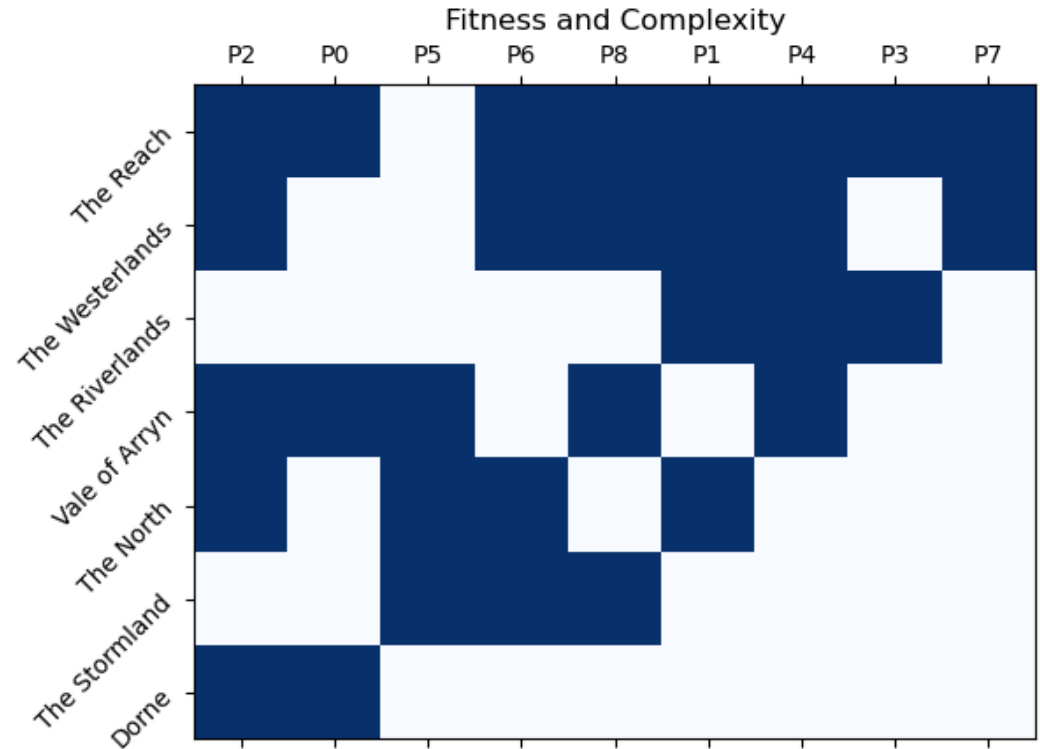
PART 2: Economic Complexity & Fitness Measures (3)



PART 2: Economic Complexity & Fitness Measures (4)

- Fitness and Complexity (1)

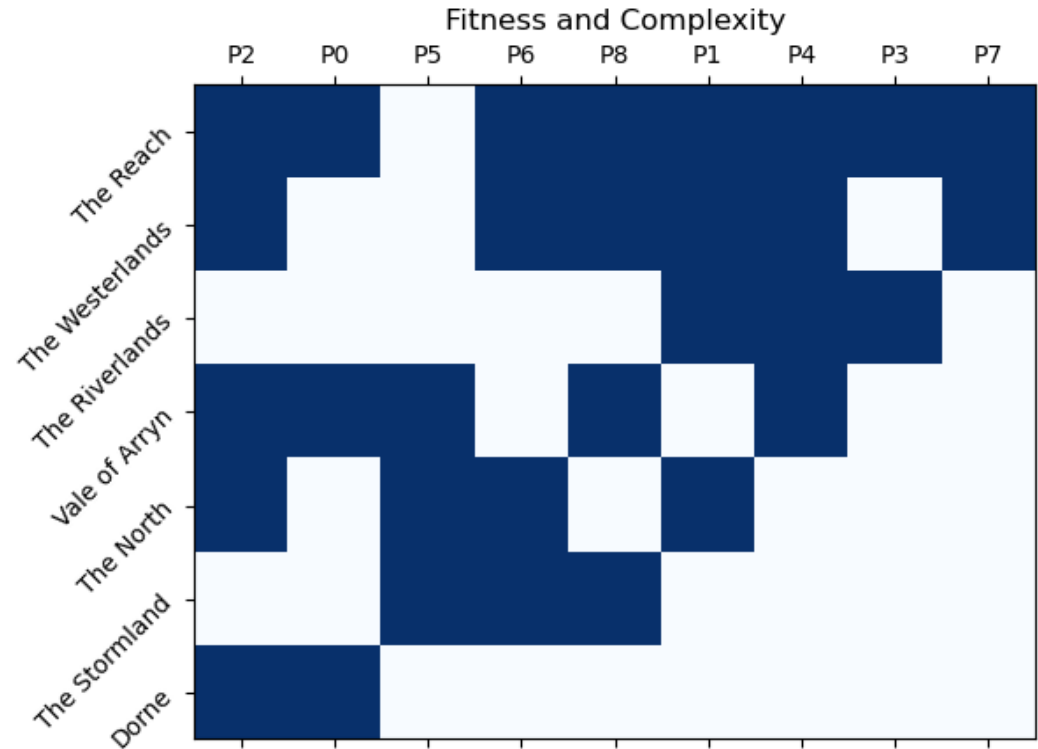
This map shows that The Reach and The Westerlands are the most developed economies with the widest activities (production). Also we can see that the **P7** and **P3** are the most rare products to make – as only two countries are able or chose to.



PART 2: Economic Complexity & Fitness Measures (5)

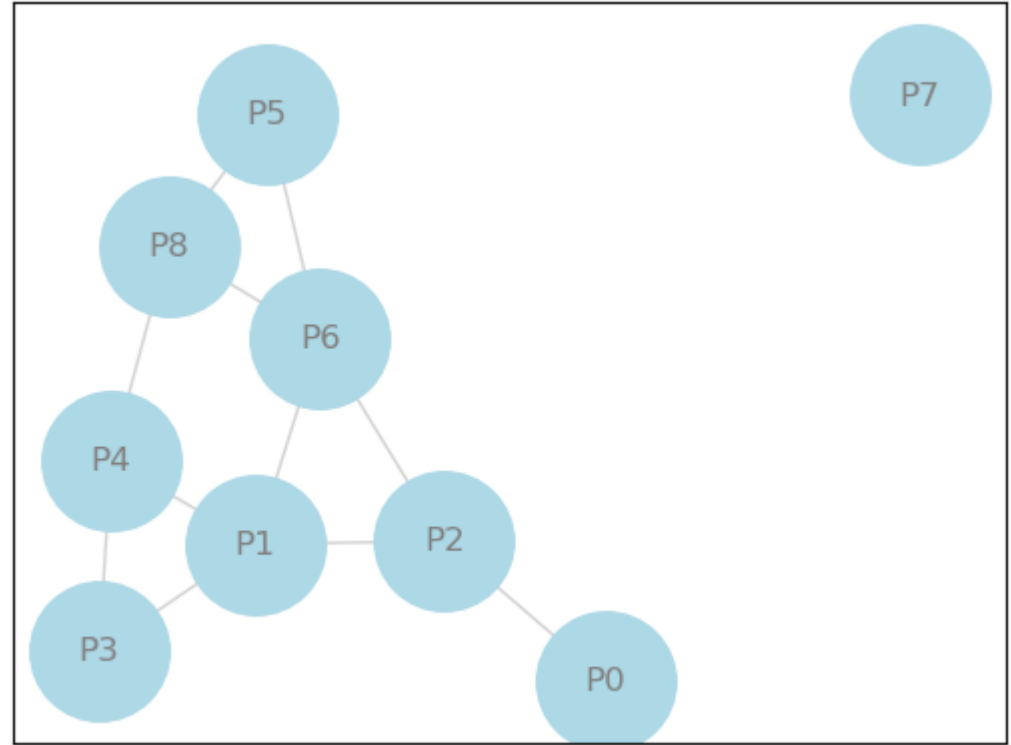
- Fitness and Complexity (2)

The Riverlands seems to specialize in top-end products of **P1**, **P4** and **P3** while country of Dorne is only capable of producing of the simplest of products **P0** and the most ubiquitous **P2**. Another specialized economy is The Stormland with only three products.



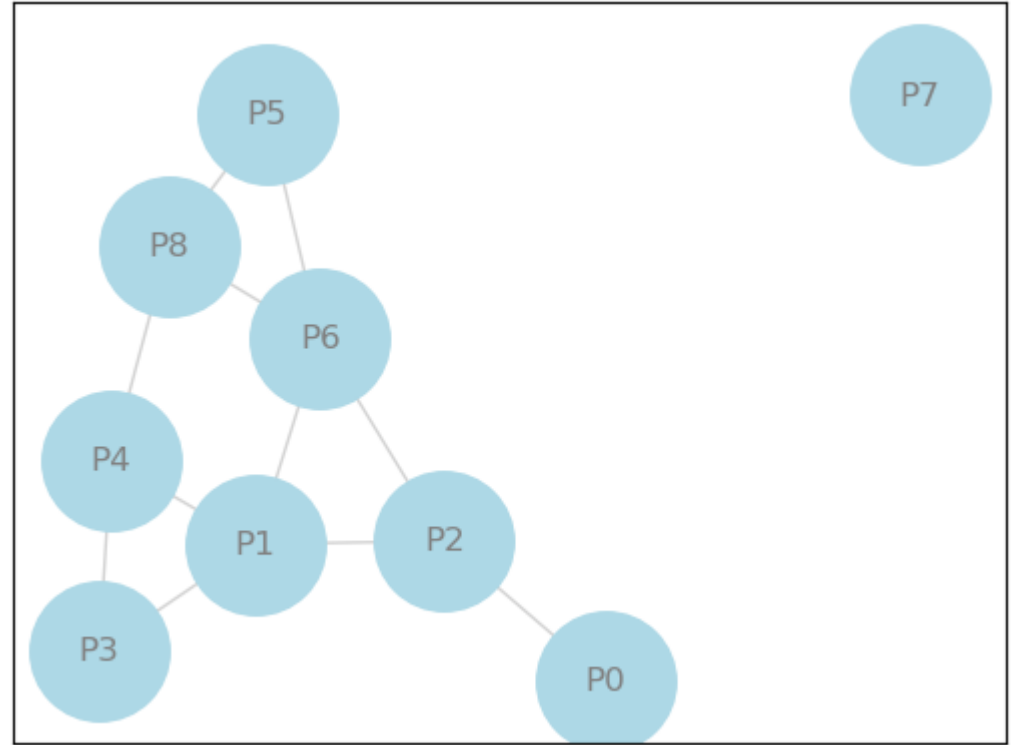
PART 3: Relatedness Between Activities (1)

- We can see that product P7 is isolated from the rest of the network and it is the least ubiquitous product to make – **only two countries developed capabilities to produce it.**



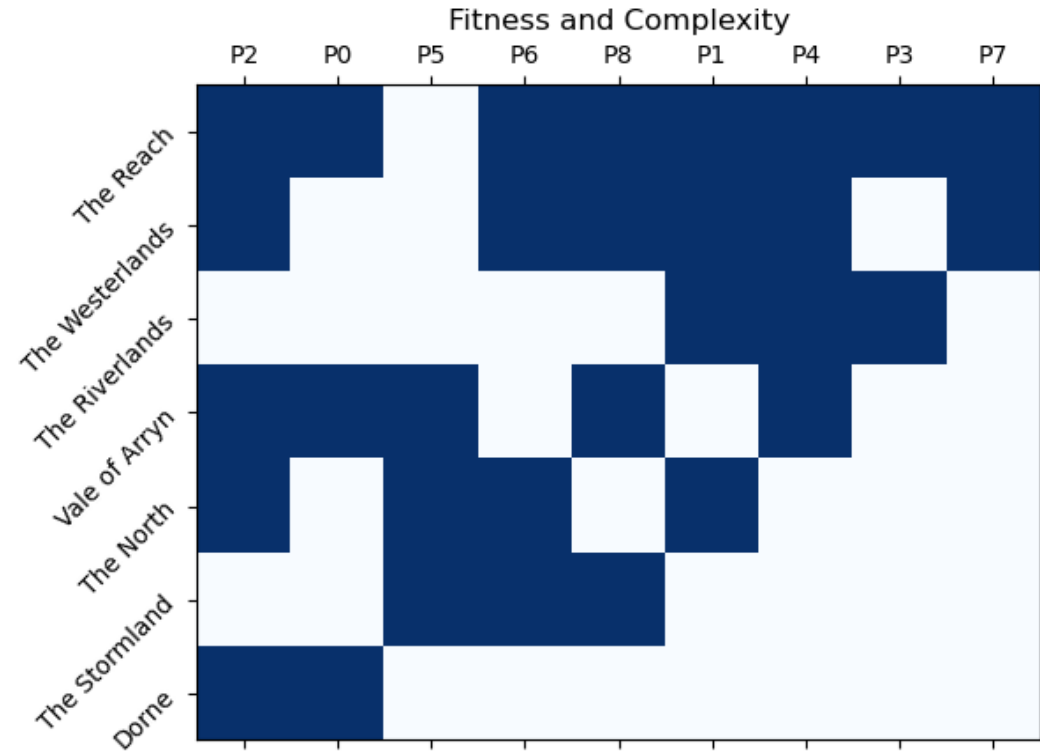
PART 3: Relatedness Between Activities (2)

- P2 is the most ubiquitous product while P1 and P6 are entry points to the more wider range of activities.
- Every country having capability of producing P2 has capability of producing P0 (next slide).

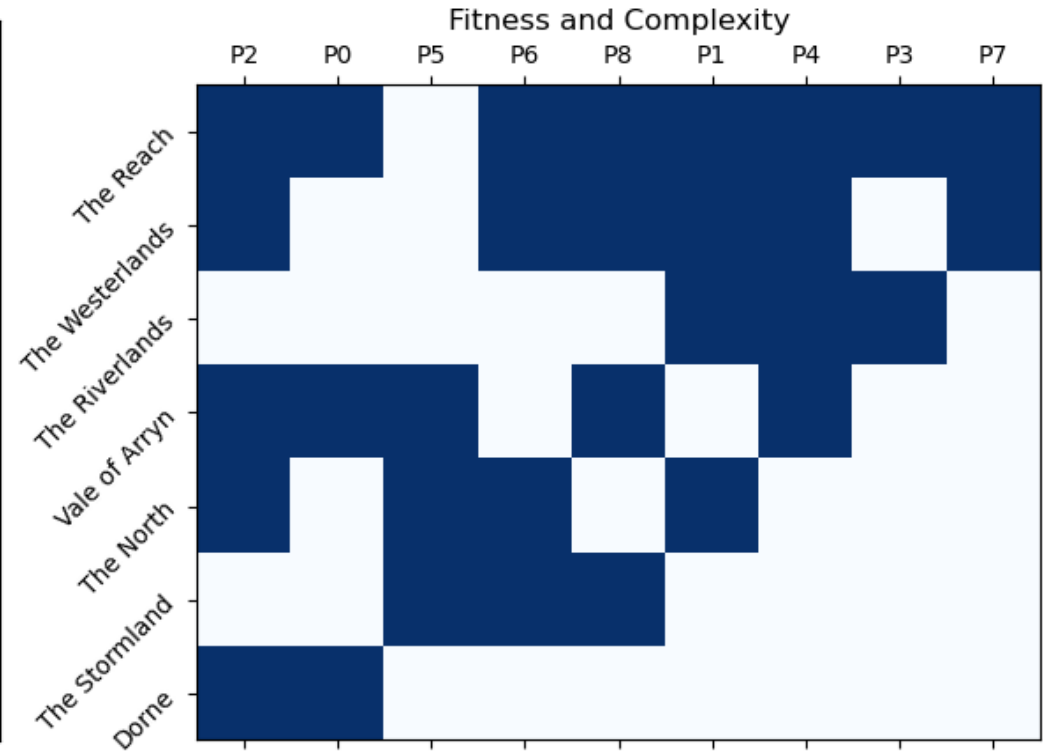
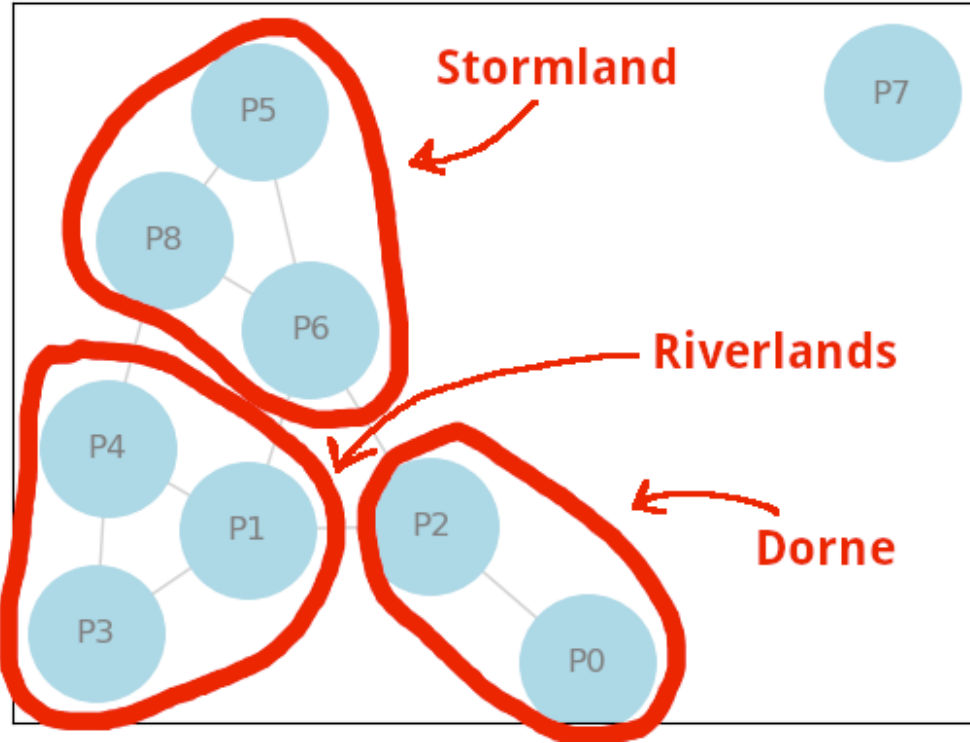


PART 3: Relatedness Between Activities (3)

- Here we can confirm that every country producing P0 is also producing P2.
- We can also see that P6 and P1 splits the map in three segments: bottom not reaching P6 capability, top having P1 capability and mid between the two.

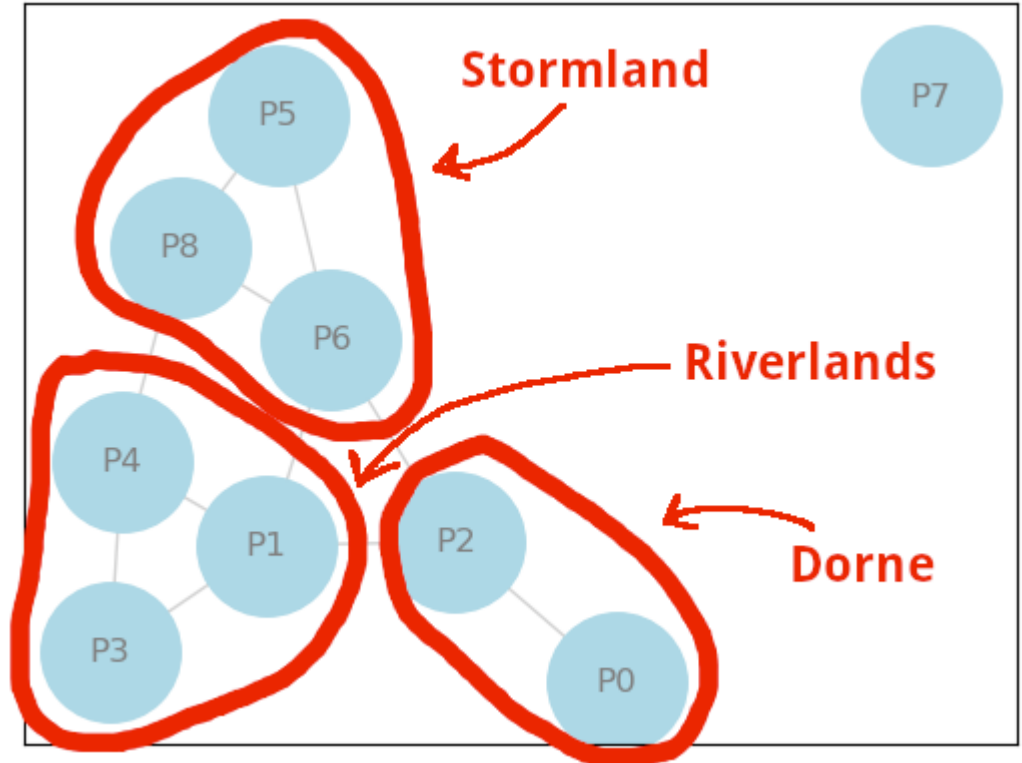


PART 4: Diversification Strategies (1)



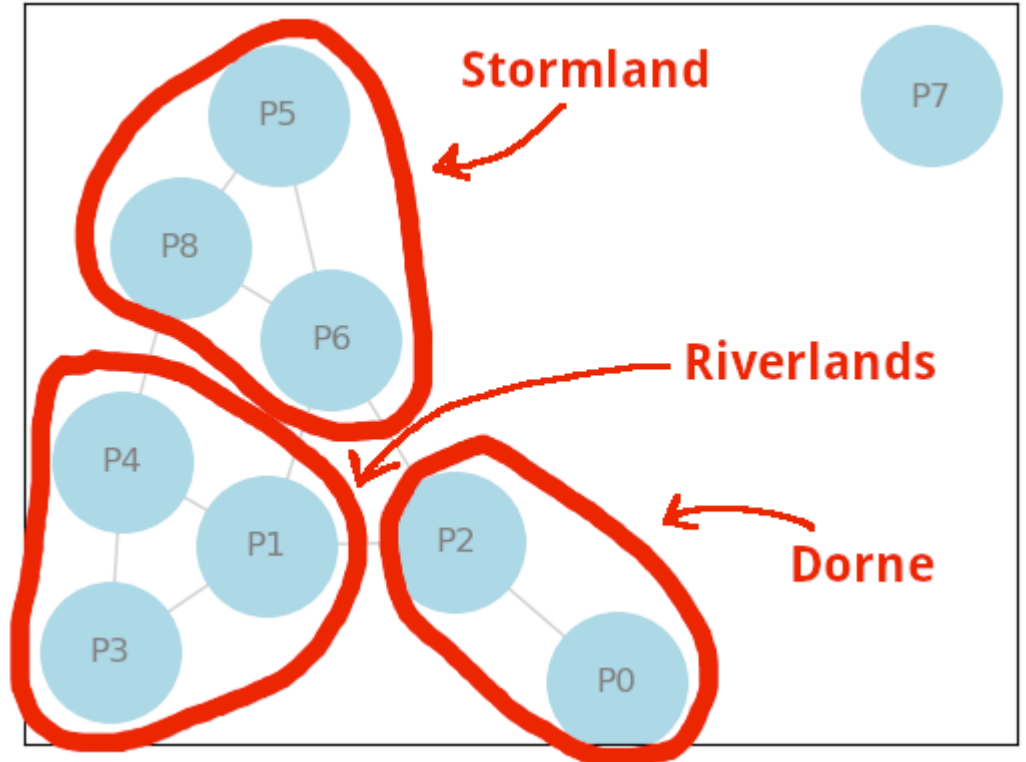
PART 4: Diversification Strategies (2)

- There is a reasonable path for **Dorne** to diversify their economy by trying to develop capability to produce product **P6** as that might open door for even more activities.



PART 4: Diversification Strategies (3)

- Similarly developing capability **P6** might help **The Riverlands** to access the more ubiquitous products of the lower segments.
- **Stormland** could on the other hand reach less ubiquitous products by trying to develop **P1** capability.



PART 5: Workflow and Contributions

- This slides and Jupyter (python) notebook is hosted on github:
 - https://github.com/ospalax/unu_merit_eci
- The collaboration started via initial email and then moved to a google's hangout chat room for offline async communication:
 - <https://chat.google.com/app/chat/AAQAdfTizyY>
 - Every group member had a gmail account and was invited



Thank You